⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 09

# Functions, filters and tests

## អនុគមន៍ការចម្រោះ និងតេស្ត

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 09

# This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្ត និងបង្កើត

## 01 · Python

Functions, filters and tests
Understand the idea and trace examples.

## 02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

## 03 · Project

Student-designed feature
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

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# Python foundations

## មូលដ្ឋាន Python

Functions, filters and tests
Concept → trace a small example → explain its behavior

ពន្យល់លក្ខណៈចម្រោះ និងតេស្តអនុគមន៍តាមការងឿ។

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# Predicates, result lists and independent tests

Predicate បញ្ជូនផ្ទាល់ និងតេស្តឯករាជ្យ

## Explicit question

Select enabled tasks that are unfinished.

## Predicate and loop

Append an ID only when both required conditions hold.

## Independent tests

Finished and disabled examples distinguish wrong filters.

Choose test inputs that distinguish the intended predicate from a wrong one.

ជ្រើស input តេស្តដែលបែងចែក predicate ត្រឹមត្រូវពីpredicate ខុស។

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# Specify behavior before changing a function

កំណត់ឥរិយាបថមុនកែអនុគមន៍

## Function contract

Name the inputs and expected returned value. Keep an ordinary case beside a boundary case.

## Filter by a predicate

A filter keeps records whose condition is True. State exactly which records should remain.

## Assertion as evidence

assert compares an observation with an expectation. Also rerun an unchanged behavior after a patch.

Understand the behavior before changing the example.

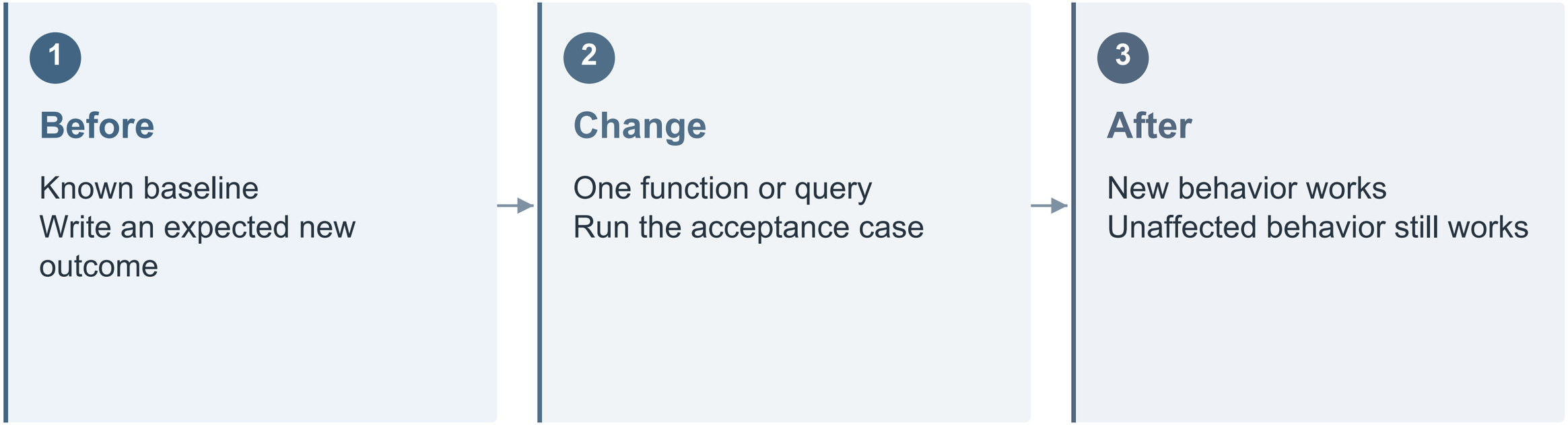
យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# A small change needs two kinds of evidence

ការកែតូចមួយត្រូវការភស្តុតាងពីរប្រភេទ




Read the diagram, then trace the same steps in code.

អានដ្យាក្រាម រួចតាមដានជំហានដដែលក្នុងកូដ។

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# Syntax to read and reuse

## វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

| Python / syntax | Meaning | When to use it |
| --- | --- | --- |
| **def feature(inputs):** | Give the change a boundary | Make behavior callable |
| **[x for x in rows if ...]** | Filter a collection | Choose the intended records |
| **assert actual == expected** | Compare against a prediction | Expose a wrong implementation |
| **check.assertEqual(a, b)** | Assert with unittest | Reuse checks in a test module |

Python Basics · sections 07, 10, 15, 17 · use only when you need a lookup.

Python Basics ផ្នែក 07, 10, 15, 17៖ មើលយោងតែពេលចាំបាច់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Check a boundary and filter a small collection

## ធ្វើប្រៃដែន និងច្រោះបណ្តុំតូច

```python
def at_least(value, minimum):
    return value >= minimum
for value in (4, 5, 6):
    print(value, at_least(value, 5))
records = [("A", False), ("B", True), ("C", False)]
pending = [name for name, finished in records if not finished]
print(pending)
```

1. Expected: 4 False; 5 True; 6 True; ['A', 'C'].
2. A predicate decides which values to keep.
3. Replace >= with >: which check should fail?

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Find a boundary bug in a function

រកកំហុសតម្លៃព្រំដែនក្នុងអនុគមន៍

```python
def before_label(portions, reviewed):
    if not reviewed:
        return "hold"
    return "team" if portions > 20 else "standard"

print(before_label(20, True))  # Wrong: standard.
print(before_label(20, False)) # Correct: hold.
```

1. Trace the early return first.
2. Compare > with >= at equality.
3. Keep an independent expected result.

An exact-boundary input can expose a one-character bug.

តម្លៃគែមព្រំដែនអាចបង្ហាញកំហុសតែមួយតួរក្បូ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Table-driven checks for a small patch

## ត្រួតពិនិត្យករណីតេស្តសម្រាប់ការកែតូចមួយ

```python
def pickup_label(portions, reviewed):
    if not reviewed:
        return "hold"
    return "team" if portions >= 20 else "standard"

cases = [(19, True, "standard"), (20, True, "team"),
         (21, True, "team"), (20, False, "hold")]
for portions, reviewed, expected in cases:
    assert pickup_label(portions, reviewed) == expected
```

1. Each tuple supplies inputs and expectation.
2. The loop runs one check per tuple.
3. The first failing assert stops execution.

A table of cases checks the patch and preserved behavior.

តារាងករណីតេស្តពិនិត្យការកែ និងឥរិយាបថដែលត្រូវរក្សា។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# List comprehensions: choose the right predicate

List comprehensions៖ ជ្រើសលក្ខខណ្ឌឱ្យត្រឹមត្រូវ

```
rows = [("D10", False, False),
        ("D11", True, False), ("D12", True, True)]
pending = [key for key, collected, delivered in rows
           if not collected]
wrong = [key for key, collected, delivered in rows
         if not delivered]
assert pending == ["D10"]
assert wrong == ["D10", "D11"]
print(pending, wrong)
```

1. Unpack each tuple into three names.
2. The if clause filters the result list.
3. Find the row that distinguishes the filters.

Changing a filter condition can change which items are included.

ប្តូរលក្ខខណ្ឌចម្រោះអាចប្តូរធាតុដែលផ្គួល។

# Import reusable code without running startup

import កូដប្រើឡើងវិញ ដោយមិនចាប់ផ្តើមកម្មវិធី

```
01 from pickup_app import planned_portions
02
03 print(planned_portions(3, 3))
04 assert planned_portions(3, 3) == 9
05 # This import does not reserve or deliver anything.
```

1. Use the documented module location.
2. Tests can call a function directly.
3. Keep input/output work in the entry point.

Reusable code can be imported without starting the application.

កូដប្រើឡើងវិញអាច import ដោយមិនចាប់ផ្តើមកម្មវិធី

# unittest: assert a returned value

unittest៖ ផ្ទៀងផ្ទាត់តម្លៃត្រឡប់

```
01 import unittest
02
03 def label(portions, reviewed):
04     if not reviewed:
05         return "hold"
06     return "team" if portions >= 20 else "standard"
07
08 check = unittest.TestCase()
09 check.assertEqual(label(20, True), "team")
10 check.assertEqual(label(20, False), "hold")
```

1. Import the actual function in your project tests
2. Retain independent expected labels
3. Add negative, failure and state-preservation cases

A test must execute an independent expectation, not remain skipped.

តេស្តត្រូវដំណើរការតាមការរំពឹងឯករាជ្យ មិនមែនរំលងទុកទេ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Hands-on practice

## លំហាត់អនុវត្តផ្ទាល់

Week_09.ipynb · Session 2
Predict → run → change → explain

ព្យាករ រត់ ប្តូរយ៉ាង និងពន្យល់លទ្ធផល។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Your turn: practise the same Python skills

ដល់វេនអ្នក អនុវត្តជំនាញ Python ដដែល

## P1–P2 · Easy

Use the approach hints. Write the code yourself in a blank cell.

## P3 · Core build

Implement a complete small program. Write expected results and assertions.

## P4–P5 · Optional

Apply the skill to a new case, then attempt a harder challenge if time permits.

Write your own code, predict the result, then test and explain it.

សរសេរកូដផ្ទាល់ ព្យាករលទ្ធផល ឆ្លុះសាក និងពន្យល់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Check your work before asking for help

ធ្វើការងារមុនសុំជួយ

## A complete implementation

Write the whole operation. Do not stop at one changed character.

## Independent expected results

Calculate a normal case and an edge case before running your program.

## A peer can explain it

Swap driver and reviewer. Explain one line and one test.

Explain a filtering condition and test a small function against an expectation.

ពន្យល់លក្ខខណ្ឌចម្រោះ និងតេស្តអនុគមន៍តាមការរំពឹង។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Project studio

## វគ្គនៃវគ្គម្រោង

Student-designed feature
Short design briefing → most time for your work

បង្កើតមុខងារតូចដែលអ្នកប្រើស្នាការ បើយតែស្គាមួយម៉ាត់

# The people’s task this week

## កិច្ចការរបស់អ្នកប្រើនាសប្តាហ៍នេះ

1

### Someone needs help

Choose one small problem a volunteer actually faces.

2

### This week, you will

Implement your approved small change and check it with another learner.

3

### A useful result

A volunteer can try the changed feature and report what happened.

Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Turn the need into an acceptance experiment

## បង្ហែតម្រូវការទៅតេស្តទទួលយក

1

### Need

Identify the coordinator decision and present difficulty.

2

### Contract

Specify inputs, permitted action and observable output.

3

### Cases

Predict a normal case and a distinguishing boundary or failure.

4

### Evidence

Run independently and compare actual records and outputs.

Expected behavior comes from the need, not from the program output.

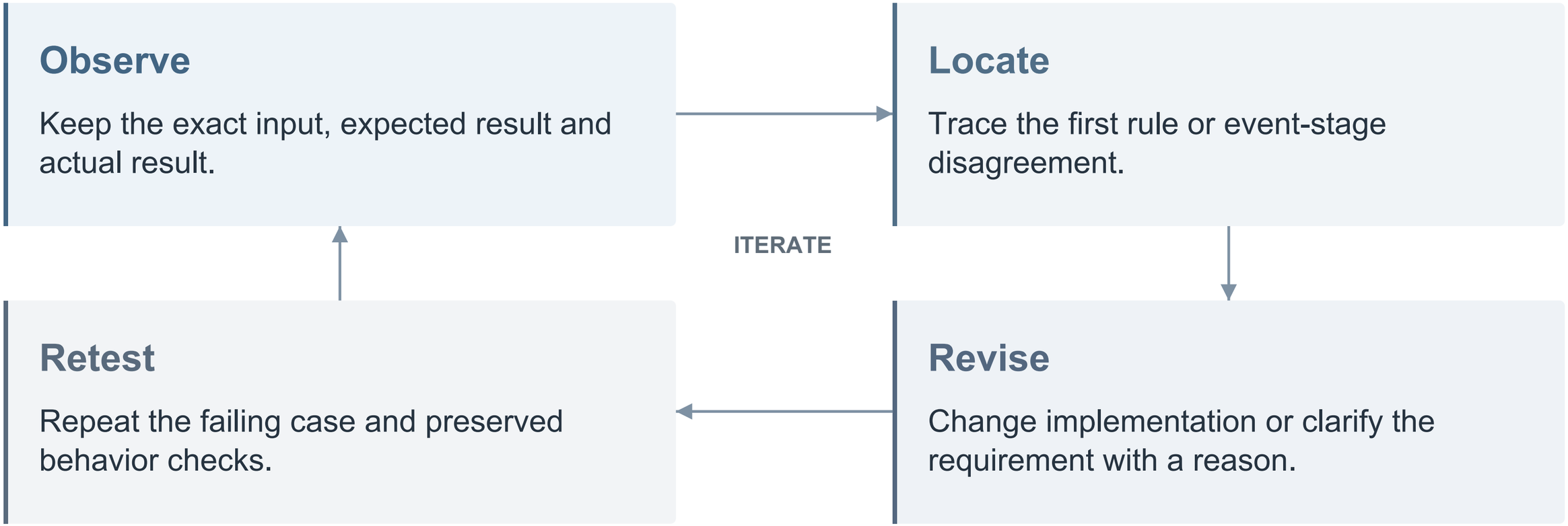
ឥរិយាបថរំពឹងចេញពីតម្រូវការ មិនមែនលទ្ធផលកម្មវិធីទេ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Use a failing observation to guide a revision

ប្រើបន្ទាល់ខុសដើម្បីណែនាំការកែ




Find the first disagreement before changing the expected result.

រកចំណុចខុសគ្នាដំបូង មុនប្តូរបន្ទាល់ផ្សេង។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Your implementation TODO

## កិច្ចការដែលអ្នកត្រូវសរសេរក្នុង

### Supplied starting point

Baseline, fixtures and sample tests are supplied; they are the starting point, not your finished feature.

### You implement / do

State the user need; implement the feature; write two acceptance cases plus an unaffected regression and a boundary/failure case.

### Check the evidence

Run your actual feature entry point. Record a real peer finding, revise and retest. For a pending-collection view, distinguish collection from delivery.

Use the supplied baseline; explain a design decision and a limitation.

ប្រើដែនគោលដែលផ្តល់; ពន្យល់ការសម្រេចរចនា និងដែនកំណត់មួយ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ



# Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

| Step | What to show |
| --- | --- |
| **1 · Locate** | Identify the input, this component and its next user. |
| **2 · Implement** | Implement your approved feature and run actual tests. |
| **3 · Check** | Two acceptance cases; regression; failure; peer retest. |
| **4 · Explain** | Show your code, its actual result and why it behaves that way. |

Use the supplied baseline; explain a design decision and a limitation.

ប្រើដែនគោលដែលផ្តល់; ពន្យល់ការសម្រេចរចនា និងដែនកំណត់មួយ។

# Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

## 1 Python unittest documentation

https://docs.python.org/3/library/unittest.html

## 2 Google Engineering Practices: How to do a code review

https://google.github.io/eng-practices/review/reviewer/

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយ យកឧទាហរណ៍មួយ និងសំណួរមួយមកជាមួយ។